

Assignment 8 - Unsupervised Learning on Titanic Dataset

Student Information

Name: Abhiney Kumar

Enrollment No.: 198409

Department: Electronics and Communication Engineering (ECE)

Institute: Dr. B.R. Ambedkar National Institute of Technology, Jalandhar

Exercise 1 - Unsupervised Data Exploration

Objective: Explore the Titanic dataset by removing the target column and performing exploratory data analysis before clustering.

```
In [1]: # =====
# Assignment 8 - Exercise 2
# K-Means Clustering on Titanic Dataset
# =====

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans

# -----
# Step 1 : Load Dataset
# -----

titanic = pd.read_csv(r"C:\Users\91953\titanic.csv")

print("***60")
print("Original Dataset")
print("***60")
print(titanic.head())

# -----
# Step 2 : Select Features
# -----

X = titanic[['Age', 'Fare', 'Pclass']]

# Fill missing values if any
X = X.fillna(X.mean())

# -----
# Step 3 : Apply K-Means
# -----

kmeans = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=10
)

titanic['Cluster'] = kmeans.fit_predict(X)

print("\nCluster Centers")
print(kmeans.cluster_centers_)

# -----
# Step 4 : Visualize Clusters
# -----

plt.figure(figsize=(8,6))

plt.scatter(
    titanic['Age'],
    titanic['Fare'],
    c=titanic['Cluster'],
    s=40
)
```

```

print(pd.isnull(train_data).sum())
=====
Original Dataset
=====
   PassengerId  Survived  Age   Sex   SibSp  Parch  Fare   Embarked
0            1         0  22.0    M     1     0    53.1000     S
1            2         1  38.0    M   29.0000     C    51.0000     S
2            3         0  26.0    M   16.0000     S    21.0000     P
3            4         1  35.0    M   16.0000     C    53.0000     S
4            5         0  35.0    M   16.0000     S    51.0000     S

   zero.4  ...  zero.12  zero.13  zero.14  Pclass  zero.15  zero.16  Embarked  \
0      0  ...      0      0      0      3      0      0      S
1      0  ...      0      0      0      1      0      0      S
2      0  ...      0      0      0      3      0      0      P
3      0  ...      0      0      0      1      0      0      S
4      0  ...      0      0      0      3      0      0      S

   zero.17  zero.18  Survived
0         0         0         0
1         0         0         1
2         0         0         0
3         0         0         1
4         0         0         0

[5 rows x 28 columns]
C:\Users\91963\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=6.
  warnings.warn(
Cluster Centers
[[ 28.98819025  15.38389418  1.54355197]
 [ 36.98684211 266.68671684  1.         ]
 [ 35.49472171  23.88536141  1.34388344]]

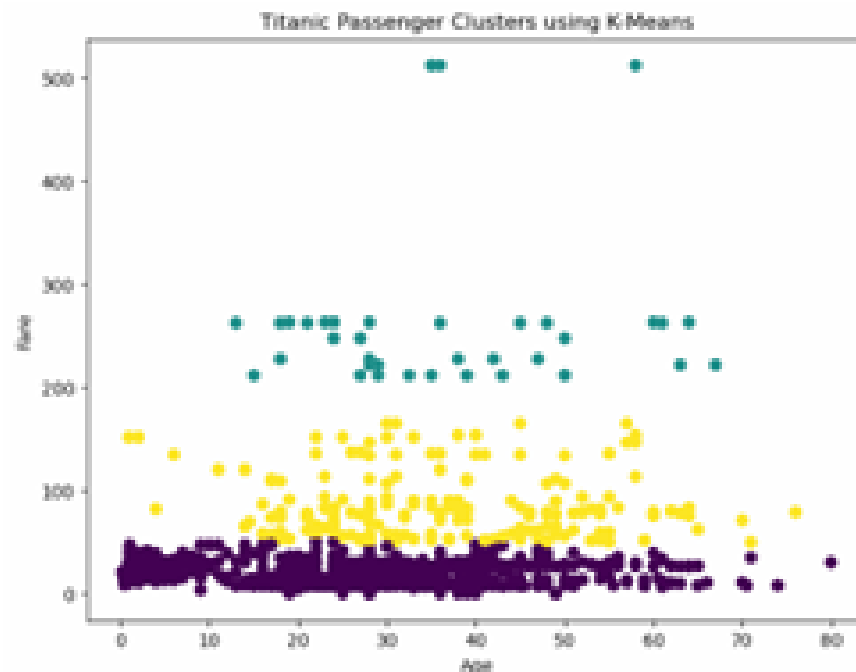
```

nv91832DownloadsExercise-02-KMeans-Clustering.ipynb.html

20

1 PM

Exercise-02-KMeans-Clustering.ipynb



```

Cluster Counts
Cluster
0      1965
1       285
2        38
Name: count, dtype: int64

```

```

Observations
-----
1. Passengers are divided into three clusters.
2. Clustering is based on Age, Fare and Passenger Class.
3. High fare passengers tend to form a separate cluster.
4. K-Means groups similar passengers without using survival labels.

```

Exercise 2 - K-Means Clustering

Objective: Apply K-Means clustering using Age, Fare and Passenger Class features and analyze the resulting clusters.

```
In [2]: # =====
# Assignment 8 - Exercise 1
# Unsupervised Learning on Titanic Dataset
# Data Exploration and Visualization
# =====

import pandas as pd
import matplotlib.pyplot as plt

# -----
# Step 1 : Load Dataset
# -----

titanic = pd.read_csv(r"C:\Users\91953\titanic.csv")

print("\n" + "="*60)
print("Original Dataset")
print("\n" + "="*60)
print(titanic.head())

# -----
# Step 2 : Remove Target Column
# -----

if 'Survived' in titanic.columns:
    unsupervised_data = titanic.drop(columns=['Survived'])
else:
    unsupervised_data = titanic.copy()

print("\n" + "="*60)
print("Dataset After Removing 'Survived' Column")
print("\n" + "="*60)
print(unsupervised_data.head())

# -----
# Step 3 : Dataset Information
# -----

print("\n" + "="*60)
print("Shape of Dataset")
print("\n" + "="*60)
print(unsupervised_data.shape)

print("\n" + "="*60)
print("Column Names")
print("\n" + "="*60)
print(unsupervised_data.columns.tolist())

print("\n" + "="*60)
print("Missing Values")
print("\n" + "="*60)
print(unsupervised_data.isnull().sum())

# -----
# Step 4 : Age Distribution
# -----
```

Original Dataset

	Passengerid	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	\
0	1	22.0	7.2500	0	1	0	0	0	0	
1	2	38.0	71.2833	1	1	0	0	0	0	
2	3	26.0	7.9250	1	0	0	0	0	0	
3	4	35.0	53.1000	1	1	0	0	0	0	
4	5	35.0	8.0500	0	0	0	0	0	0	

	zero.4	...	zero.12	zero.13	zero.14	Pclass	zero.15	zero.16	Embarked	\
0	0	...	0	0	0	3	0	0	2.0	
1	0	...	0	0	0	1	0	0	0.0	
2	0	...	0	0	0	3	0	0	2.0	
3	0	...	0	0	0	1	0	0	2.0	
4	0	...	0	0	0	3	0	0	2.0	

	zero.17	zero.18	Survived
0	0	0	0
1	0	0	1
2	0	0	1
3	0	0	1
4	0	0	0

[5 rows x 28 columns]

Dataset After Removing 'Survived' Column

	Passengerid	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	\
0	1	22.0	7.2500	0	1	0	0	0	0	
1	2	38.0	71.2833	1	1	0	0	0	0	
2	3	26.0	7.9250	1	0	0	0	0	0	
3	4	35.0	53.1000	1	1	0	0	0	0	
4	5	35.0	8.0500	0	0	0	0	0	0	

	zero.4	...	zero.11	zero.12	zero.13	zero.14	Pclass	zero.15	zero.16	\
0	0	...	0	0	0	0	3	0	0	
1	0	...	0	0	0	0	1	0	0	
2	0	...	0	0	0	0	3	0	0	
3	0	...	0	0	0	0	1	0	0	
4	0	...	0	0	0	0	3	0	0	

	Embarked	zero.17	zero.18
0	2.0	0	0
1	0.0	0	0
2	2.0	0	0
3	2.0	0	0
4	2.0	0	0

[5 rows x 27 columns]

Shape of Dataset

(1389, 27)

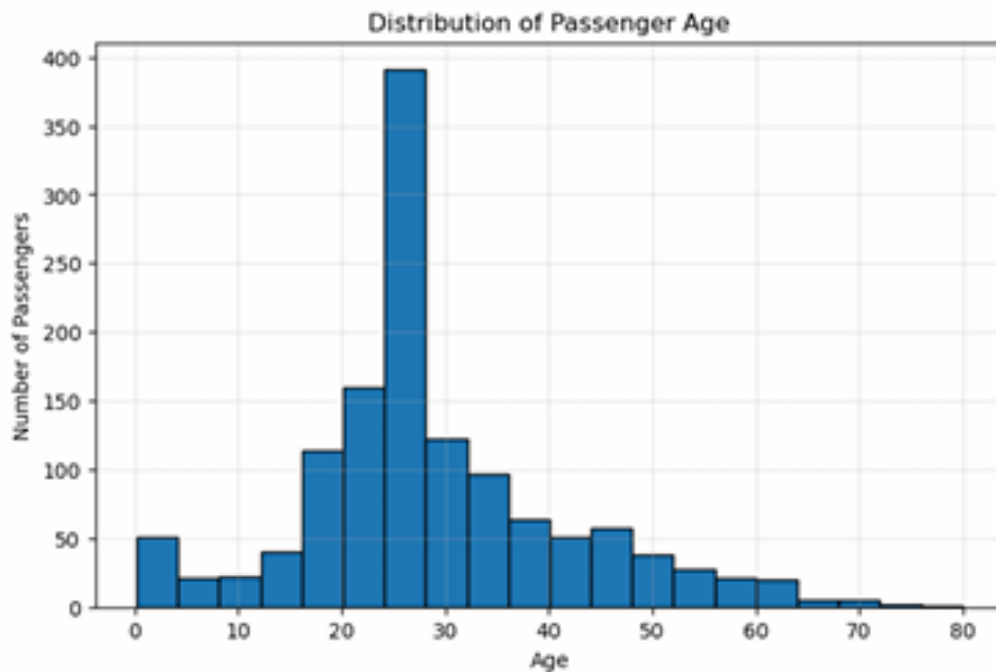
Column Names

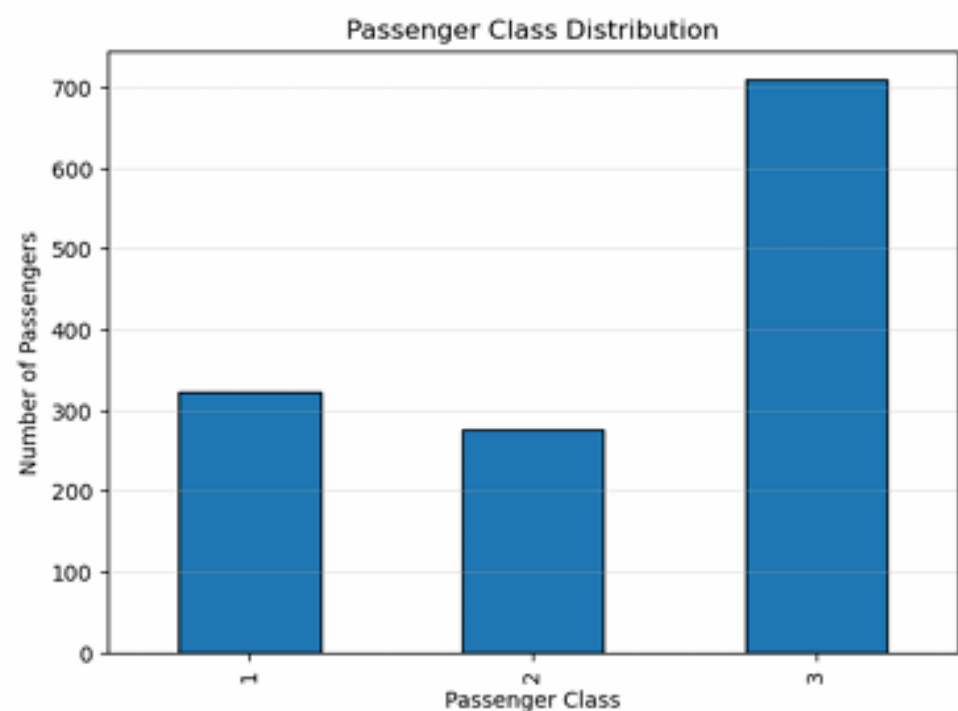
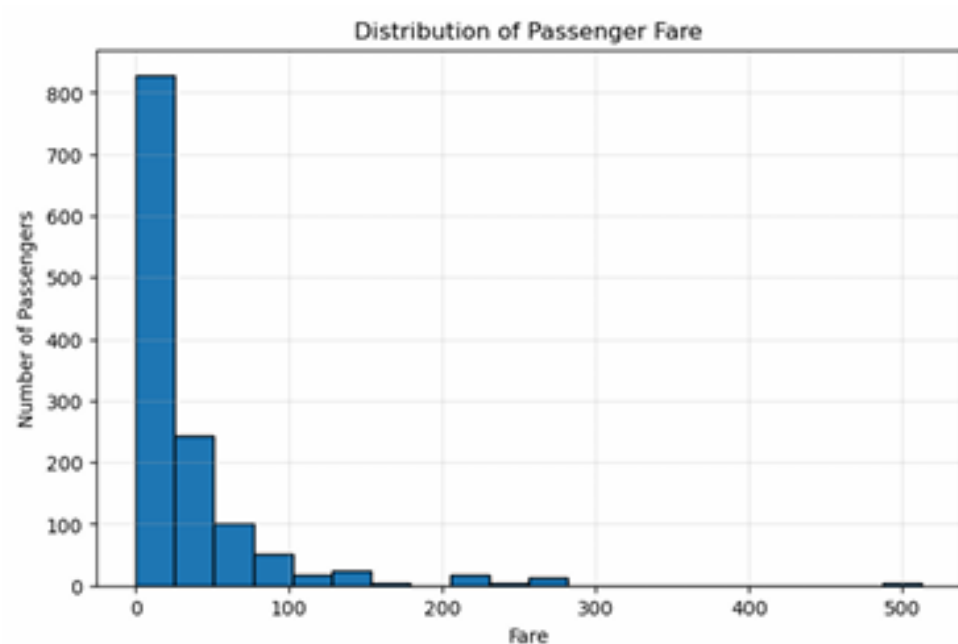
```
['Passengerid', 'Age', 'Fare', 'Sex', 'sibsp', 'zero', 'zero.1', 'zero.2', 'zero.3', 'zero.4', 'zero.5', 'zero.6', 'Parch', 'zero.7', 'zero.8', 'zero.9', 'zero.10', 'zero.11', 'zero.12', 'zero.13', 'zero.14', 'Pclass', 'zero.15', 'zero.16', 'Embarked', 'zero.17', 'zero.18']
```

```
=====
Missing Values
=====
```

```
Passengerid    0
Age             0
Fare           0
Sex            0
sibsp          0
zero           0
zero.1         0
zero.2         0
zero.3         0
zero.4         0
zero.5         0
zero.6         0
Parch          0
zero.7         0
zero.8         0
zero.9         0
zero.10        0
zero.11        0
zero.12        0
zero.13        0
zero.14        0
Pclass         0
zero.15        0
zero.16        0
Embarked       2
zero.17        0
zero.18        0
```

```
dtype: int64
```





```
=====
Observations
=====
```

1. Most passengers are adults between 20 and 40 years of age.
2. Passenger fares are right-skewed, with most passengers paying lower fares.
3. Passenger Class 3 has the highest number of passengers.
4. The dataset is now ready for Unsupervised Learning algorithms such as K-Means Clustering.

Exercise 1 Completed Successfully.

Exercise 3 - Reflection Questions

Q1. Why is the target column removed in Unsupervised Learning?

Unsupervised learning does not use output labels. The algorithm discovers hidden patterns directly from the input features.

Q2. What is the role of K-Means Clustering?

K-Means groups similar observations into clusters based on feature similarity.

Q3. What did you learn from this assignment?

I learned how to explore datasets, preprocess data, visualize distributions, apply K-Means clustering and interpret cluster results without target labels.